AGRO-ECOLOGY AND AGRO-BIODIVERSITY COURSE

An Agro-ecology course typically covers sustainable agricultural practices, soil health, biodiversity, and social-ecological systems to create resilient food systems. Key topics include ecological principles, soil management, pest control, Agroforestry, and socioeconomic aspects like food sovereignty and market access, suitable for interns, students, farmers, and development practitioners.

Typical Course Curriculum

- **Ecological Principles**
  - Introduction to Agro-ecology and biodiversity
  - Principles of Agro-ecology and biodiversity
  - The core components of Agro-ecology

- ## Soil Management
  - Land preparation techniques
  - Soil aeration
  - Types of seed bed preparation
  - Composting and soil nutrition: Techniques for making, managing, and applying compost to build long-term soil health using local materials

- ## Pest Control
  - Natural pest control using beneficial insects
  - Intercropping
  - Organic remedies

- ## Agroforestry and Apiculture
  - Introduction to Agroforestry, importance to Kenya's social economy
  - Nursery types, in Agroforestry
  - Principles and importance of tree planting in Agroforestry
  - Agroforestry practices and systems – hedgerows, alleys/ spacing, cropping, stripping, rotational fallow
  - Selection and management of tree species in Agroforestry
  - Bee-keeping: Apiculture: Behavior, biology and nutrition of honey bees; Beehive types; traditional and modern; setting an apiary; Beekeeping equipment; Harvesting, processing and utilization of products; Pests and diseases of bees and their control.

- ## Food Sovereignty: Family Nutrition and Health
  - Introduction to the principles of human nutrition and world food problem with

emphasis on East Africa

- Nutrients and their sources: Energy requirements and metabolism; Vitamins and minerals; Functions of nutrients in the human body; proteins; Carbohydrates; Fats.
- Nutrients in the balanced diet.
- Malnutrition in children: symptoms, causes and cure.
- Different assessment (Methods) of nutritional status.
- Foods, preparation, preservation, utilization and storage.
- Introduction to the use of food consumption surveys, tables and nutritional status survey
- Family life education.

• Market

- Economics and Marketing: Record keeping, business planning, farm finance,
- Marketing and value addition of animal products.
- Use of farm management software and computer applications.

Key Learning Outcomes

- An integrated/holistic understanding of food systems based on Agro-ecology/Agro-biodiversity principles and practices to contribute to Agro-ecological transformation.
- A community of practice established assuring a long –term implementation of Agro-ecology.
- Catalytic relations and powerful initiatives developed through well-coordinated networks of alumni

Typical Course curriculum

- **Duration:** At one week, two weeks, one month, three months, nine months, one-year modular courses
- **Assessment:** Mix of lectures, laboratory experiments, field studies, participatory on-farm demonstrations, on-farm research projects, and examinations. Field work involves practical and experiential learning of different Agro-ecological principles and practices. This course can be carried out through e-learning platform, face-to-face sessions, and field work
- **Requirements:** Background in biology, chemistry, mathematics, or geography.